space and time that for many of us are second nature have turned out to be figments of a false Newtonian perspective.

Quantum Reality

The second anomaly to which Lord Kelvin referred led to the quantum revolution, one of the greatest upheavals to which modern human understanding has ever been subjected. By the time the fires subsided and the smoke cleared, the veneer of classical physics had been singed off the newly emerging framework of quantum reality.

A core feature of classical physics is that if you know the positions and velocities of all objects at a particular moment, Newton's equations, together with their Maxwellian updating, can tell you their positions and velocities at any other moment, past or future. Without equivocation, classical physics declares that the past and future are etched into the present. This feature is also shared by both special and general relativity. Although the relativistic concepts of past and future are subtler than their familiar classical counterparts (Chapters 3 and 5), the equations of relativity, together with a complete assessment of the present, determine them just as completely.

By the 1930s, however, physicists were forced to introduce a whole new conceptual schema called *quantum mechanics.* Quite unexpectedly, they found that only quantum laws were capable of resolving a host of puzzles and explaining a variety of data newly acquired from the atomic and subatomic realm. But according to the quantum laws, even if you make the most perfect measurements possible of how things are today, the best you can ever hope to do is predict the *probability* that things will be one way or another at some chosen time in the future, or that things were one way or another at some chosen time in the past. The universe, according to quantum mechanics, is *not* etched into the present; the universe, according to quantum mechanics, participates in a game of chance.

Although there is still controversy over precisely how these developments should be interpreted, most physicists agree that probability is deeply woven into the fabric of quantum reality. Whereas